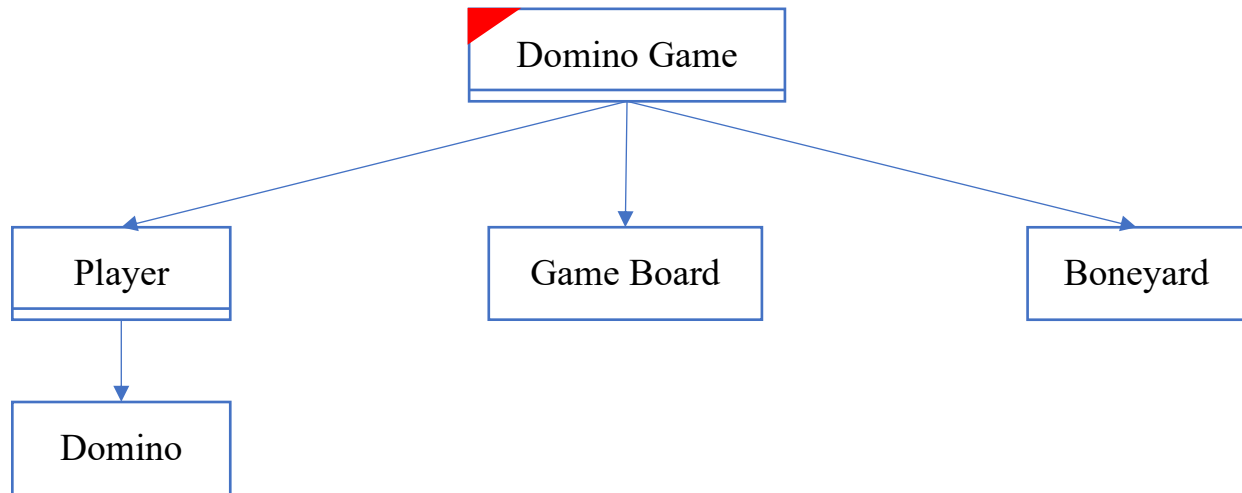


Proposed Design of Version 1



Trigger: update from keyboard

Description of the Design

Domino Game : Central manager of the game flow and state

- Data:
 - human: Player object representing the human player.
 - computer: Player object representing the computer player.
 - boneyard: Boneyard object.
 - board: GameBoard object.
 - scanner: Scanner object for user input.
 - lastPlayer: Player object representing the last player to make a move
- Behavior:
 - DominoGame(): Initialize players, boneyard, game board and scanner
 - play(): Start and manages the main game.
 - dealInitialHands(): Distributes initial dominos to players.
 - humanTurn(): Manages human player's turn.
 - computerTurn(): Manages computer player's turn.
 - playHumanDomino(): Handles the human player's process of choosing and playing a domino.
 - playComputerDomino(): Handles the computer player's process of choosing and playing a domino.
 - hasPlayableDomino(Player): Checks if a player has a playable domino.
 - canPlay(Domino): Checks if a domino can be played.

- canPlayLeft(Domino): Checks if a domino can be played on the left end of the board
- canPlayRight(Domino): Checks if a domino can be played on the right end of the board.
- isValidPlay(Domino, boolean): Checks if playing a domino is valid.
- isGameOver(): Checks if the game is over.
- displayGameState(): Displays the current state of the game.
- announceWinner(): Announces the winner of the game.

GameBoard: Represents the playing area where dominos are placed.

- Data:
 - playedDominos: List of Domino objects on the board.
 - leftEnd: int representing the value on the left end of the board.
 - rightEnd: int representing the value on the right end of the board.
- Behavior:
 - playDomino(Domino, boolean): Plays a domino to the board.
 - getLeftEnd(): Gets the value at the left end of the board.
 - getRightEnd(): Gets the value at the right end of the board.
 - toString(): Provides a string representation of the game board.

Player: Represents a player in the game.

- Data:
 - hand: List of Domino objects.
 - name: String representing the player's name.
- Behavior:
 - addDomino(Domino): Adds a domino to the player's hand.
 - getHand(): Gets the player's current hand.
 - getScore(): Calculates the player's score based on the sum of dominos in their hand.
 - removeDomino(Domino): Removes a domino from the player's hand.

Boneyard: Represents the pool of undrawn dominos.

- Data:
 - dominos: List of Domino objects not yet in play.
- Behavior:
 - draw(): Draws a domino from the boneyard.
 - isEmpty(): Checks if the boneyard is empty.
 - size(): Gets the current size of the boneyard.

Domino: Represents a single domino piece.

- Data:
 - left: int representing the value on the left side.
 - right: int representing the value on the right side
- Behavior:
 - getLeft(): Gets the left value of the domino.
 - getRight(): Gets the right value of the domino.
 - rotate(): Rotates the domino by swapping its left and right values.
 - getSum(): Calculates the sum of the domino's values
 - toString(): Provides a string representation of the domino.